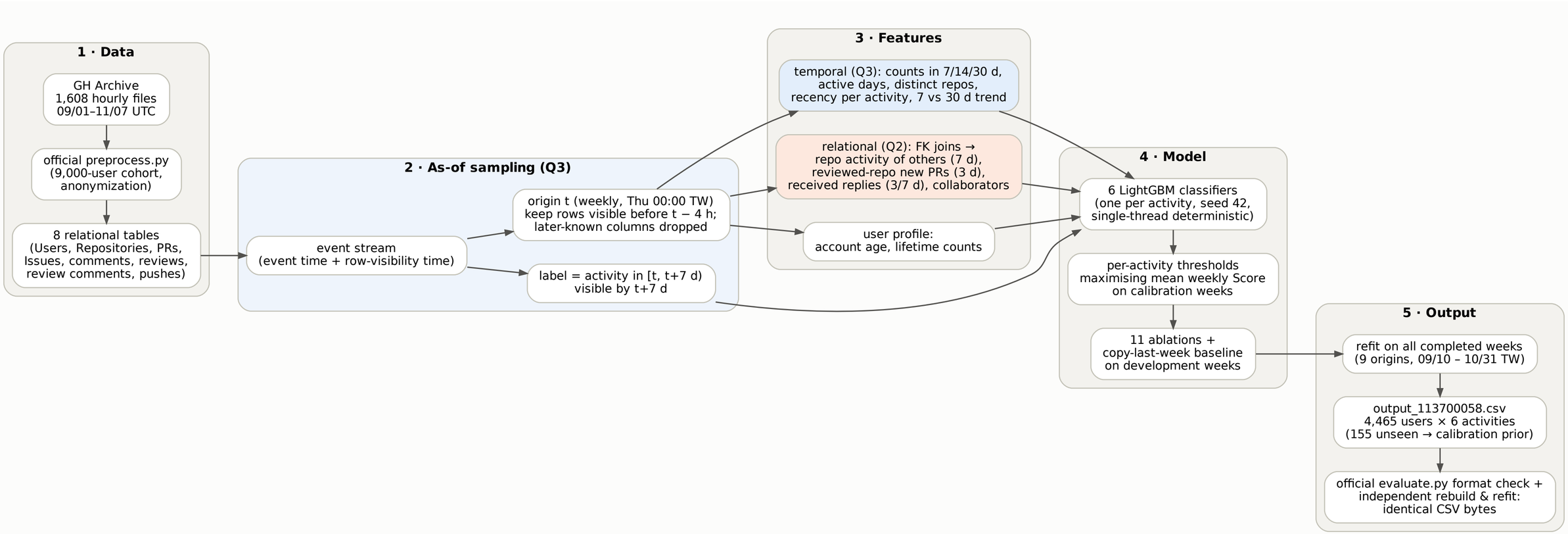
1 · Data
GH Archive
1,608 hourly files
09/01–11/07 UTC
official preprocess.py
(9,000-user cohort,
anonymization)
8 relational tables
(Users, Repositories, PRs,
Issues, comments, reviews,
review comments, pushes)
2 · As-of sampling (Q3)
event stream
(event time + row-visibility time)
origin t (weekly, Thu 00:00 TW)
keep rows visible before t − 4 h;
later-known columns dropped
label = activity in [t, t+7 d)
visible by t+7 d
3 · Features
temporal (Q3): counts in 7/14/30 d,
active days, distinct repos,
recency per activity, 7 vs 30 d trend
relational (Q2): FK joins →
repo activity of others (7 d),
reviewed-repo new PRs (3 d),
received replies (3/7 d), collaborators
user profile:
account age, lifetime counts
4 · Model
6 LightGBM classifiers
(one per activity, seed 42,
single-thread deterministic)
per-activity thresholds
maximising mean weekly Score
on calibration weeks
11 ablations +
copy-last-week baseline
on development weeks
5 · Output
refit on all completed weeks
(9 origins, 09/10 – 10/31 TW)
output_113700058.csv
4,465 users × 6 activities
(155 unseen → calibration prior)
official evaluate.py format check +
independent rebuild & refit:
identical CSV bytes